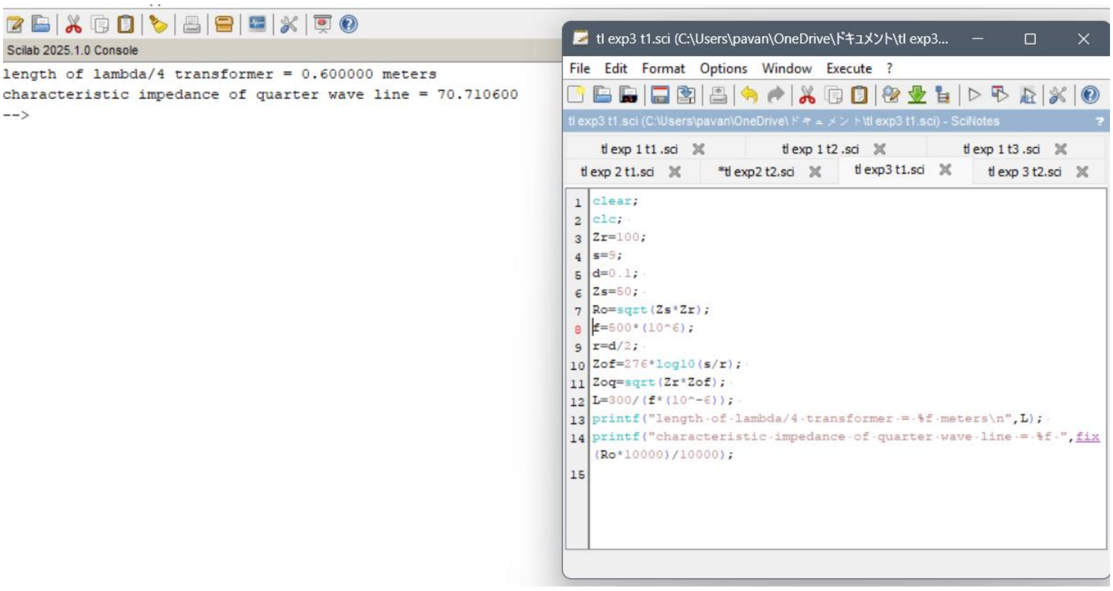
Designing a Quarter wave transformer by varying following parameters a) frequency b) input impedance c) characteristic impedance

Test case 1:

```
clear;

clc;

Zr=100;

s=9;

d=0.1;

Zs=50;

 Ro=sqrt(Zs*Zr);f=500*(10^6);

r=d/2; Zof=276*log10(s/r);

Zoq=sqrt(Zr*Zof);

 L=300/(f*(10^-6));

printf("length of lambda/4 transformer = %f meters\n",L);

printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);
```



Test case 2:

```
clear;

clc;
```

```
Zr=100;

s=9;

d=0.1;

Zs=50;

Ro=sqrt(Zs*Zr);f=100*(10^6);r=d/2;

Zof=276*log10(s/r);

Zoq=sqrt(Zr*Zof);

L=300/(f*(10^-6));

printf("length of lambda/4 transformer = %f lambda\n",L);

printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);
```